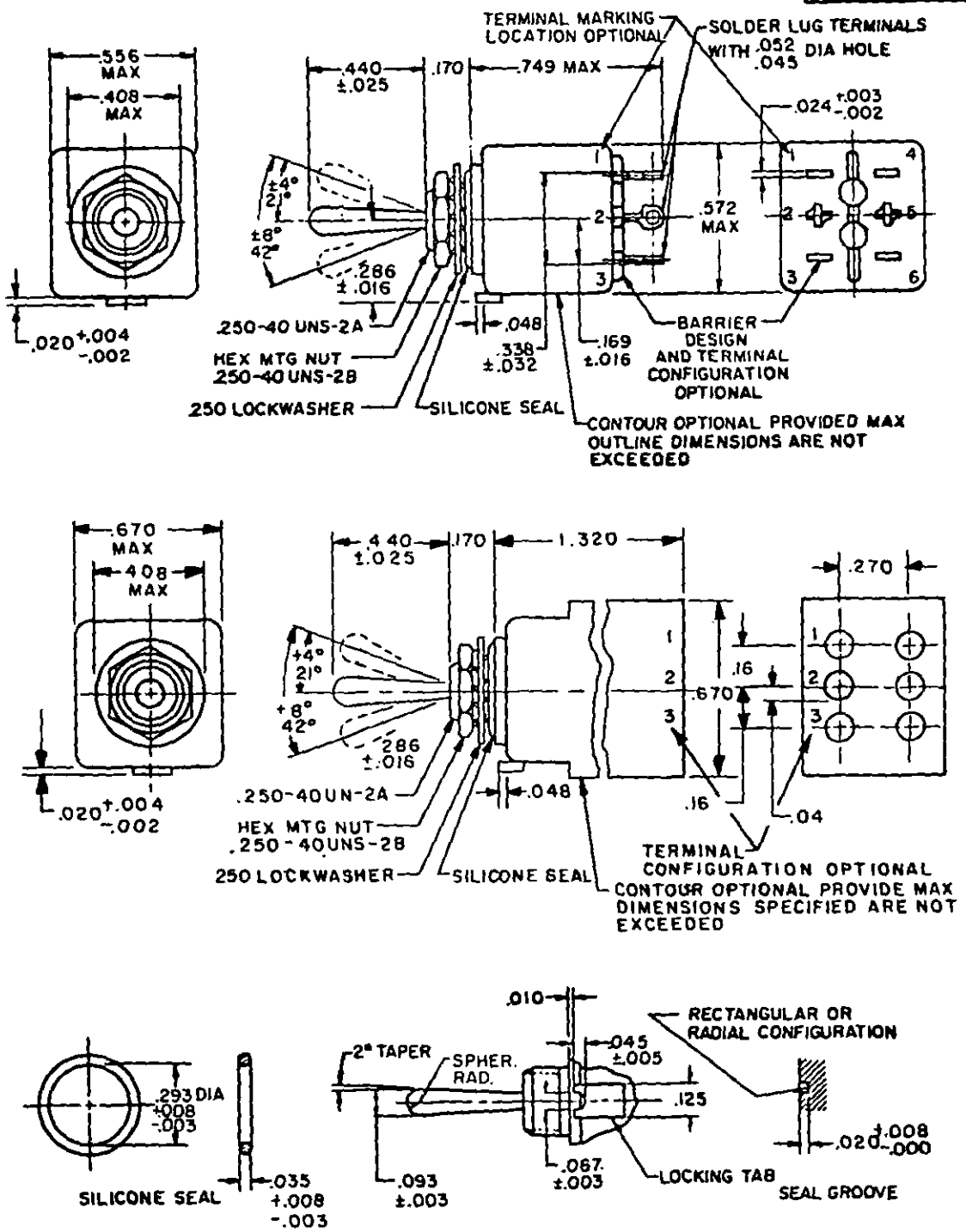
FED. SUP CLASS
5930



Ⓚ Denotes change

P.A	International interest	TITLE	MILITARY STANDARD
AF-85 Other Cust ARMY-ER NAVY-AS	ASCCAS 12/16 A	SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE TOGGLE SEALED, SINGLE POLE, SOLDER LUG, OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS	MS24656
Procurement Specification MIL-S-8834		SUPERSEDES: MS0024656 (USAF)	PAGE 1 OF 5

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Distribution Statement A. Approved for public release; distribution is unlimited.

User activities: Army - MI, Navy - , Air Force -

Review activities: Army - AR, AV, Navy - EC, Air Force - 11, 17, 99

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APPROVED 24 FEB 60 REVISED (J) 20 JULY 1984 (K) 18 OCT 1985

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Review activities: Army - AR, AV
Navy - EC
Air Force - 11, 17, 99

User activities: Army - MI
Navy -
Air Force -

DD FORM 1 MAY 73 672 (Coordinated) previous editions of this form are obsolete.

Procurement Specification: MIL-S-8834

P.A: AF-85
Other Cust: ARMY-ER
NAVY-AS

International Interest: ASCCAS 12/16A

SUPERSEDES: MS0024656 (USAF)

TITLE: SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE, TOGGLE SEALED, DOUBLE POLE, SOLDER LUG OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS

MILITARY STANDARD

MS24656

PAGE 2 OF 5

DETAIL REQUIREMENTS: 7/

MS DASH NO	CIRCUITS MADE BETWEEN TERMINALS AS INDICATED WITH THE TOGGLE LEVER IN THESE POSITIONS: 1/			CURRENT CAPACITY IN AMPERES						LIFE LOW CURRENT LEVEL SWITCHING 5 mV
				RESISTIVE LOAD			INDUCTIVE LOAD			
					115 VOLTS			115 VOLTS		
	OPPOSITE LOCKING 4/ TAB SIDE	CENTER POSITION	LOCKING TAB SIDE 4/	28 VOLTS AC	60 HERTZ AC	400 HERTZ AC	28 VOLTS DC	60 HERTZ AC	400 HERTZ AC	
-211		OFF	1-2,4-5 ON							
-221	2-3,5-6 ON	NONE	OFF							
-231			1-2,4-5 ON				2/			3/
-241			NONE	5	2	3	1	1	2	25 μA
-271	MOM-ON 2-3,5-6	OFF								
-281	NONE		MOM-ON 1-2,4-5							
-311	ON 2-3,5-6									
-321 5/6/	NONE	2-3 5-6 ON	1-2 4-5 MOM-ON							
-331 5/6/	2-3 5-6 ON	2-3 4-5 ON	1-2 4-5 ON							
-341 5/6/	2-3 5-6 MOM-ON	2-3 4-5 ON	1-2 4-5 MOM-ON							
-351 5/6/	2-3 5-6 ON	2-3 4-5 ON	1-2 4-5 MOM-ON							

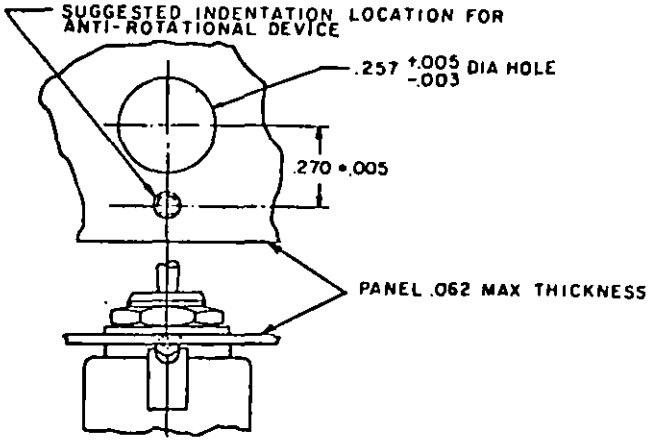
NOTES:
1/ Direction of movement of internal mechanism is opposite to the direction of the toggle movement.
2/ With time constant of .020 ±.002 seconds.
3/ Contact resistance not to exceed 50 ohms during life, low current level switching.
4/ Nonfunctional terminals shall not be supplied.
5/ Dielectric withstanding voltage; 1,200 V rms at sea level (center on circuits).
6/ Delayed action of the switch toggle lever may cause circuit to close or open before snap action mechanism trips.
7/ For IWTS termination MS number will be concluded by the letter W. Example part numbers: MS24656-711, toggle sealed on-off-on, miniature, MS24656-711W, toggle sealed, on-off-on, miniature with IWTS termination.

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INCHES	MM	INCHES	MM	INCHES	MM
.002	0.05	.032	0.81	.250	6.35
.003	0.08	.035	0.89	.257	6.53
.004	0.10	.040	1.02	.270	6.86
.005	0.13	.045	1.14	.286	7.26
.008	0.20	.048	1.22	.293	7.44
.010	0.25	.052	1.32	.338	8.59
.016	0.41	.062	1.57	.408	10.36
.020	0.51	.067	1.70	.440	11.18
.024	0.61	.093	2.36	.456	14.12
.025	0.64	.125	3.18	.572	14.53
.028	0.71	.160	4.06	.670	17.02
.032	0.81	.169	4.29	.749	19.02
		.170	4.32	1.320	33.53

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Review activities: Army - AR, AV, Navy - EC, Air Force - 11, 17, 99

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P.A	International interest	TITLE	MILITARY STANDARD
AF-85 Other Cust ARMY-ER NAVY-AS	ASCCAS 12/16A	SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE, TOGGLE SEALED, DOUBLE POLE, SOLDER LUG OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS	MS24656
Procurement Specification MIL-S-8834		SUPERSEDES: MS0024656 (USAF)	PAGE 3 OF 5

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User activities: Army - MI
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(K) NOTES:

1. Dimensions are in inches.
2. *Metric equivalents are given for general information only.*
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) on decimals and ±.5° on angles.
4. For hardware detail specifications see appendix of MIL-S-8834.
5. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.
7. Altitude 50,000 feet.
8. 115 V ac,60 hertz electrical endurance tests are to be performed at room temperature and pressure.
9. Weight: .0143 pound maximum (6.5 grams maximum) with solder lug terminals.
   .0300 pound maximum (12.6 grams maximum) with IWTS termination.
10. Shock: Switches with solder terminals, Method I and Method II (high impact). Switches with IWTS terminals, Method I only. The switch shall be electrically and mechanically operative at the conclusion of the test and there shall be no mechanical transfer during the test.
11. Suitable for mounting on panels of .062 (1.57 mm) maximum thickness.
12. Strength of terminal: 5 pound normal to the mounting planes, and 2 pounds in other planes, (solder lug terminals only).
13. Strength of actuator: Lever pivot and lever stop 6 pounds.
14. Switches with IWTS terminals shall accept M39029/1-101 connector contacts.
15. With IWTS termination, sealing plugs may be used in nonfunctional terminal grommet holes. Switches shall accept MS27488-10 end seal plugs.
16. The terminal sealing grommet for switches with IWTS terminals shall seal on smooth wire insulations of .040 (1.02 mm) to .083 (2.11 mm) diameter. The sealing grommet shall be color coded red to indicate contact size.
17. Installing/removal tool for IWTS connector contacts is M81969/14-02.

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P.A AIR FORCE - 85 Other Cust ARMY - ER NAVY - AS	International interest ASCCAS 12 / 16 A	TITLE SWITCH, TOGGLE, POSITIVE BREAK, MINIATURE, TOGGLE SEALED, DOUBLE POLE, SOLDER LUG OR INTEGRATED WIRE TERMINALS; .250 MOUNTING BUSHINGS	MILITARY STANDARD MS 24656
Procurement Specification MIL - S - 8834		SUPERSEDES: MS0024656(USAF)	PAGE 4 OF 5

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The following test requirements apply only to those switches with integrated wire terminals:

TEST REQUIREMENTS:

Qualification and group B tests are to be performed in accordance with MIL-S-8834 except as follows:

During all tests switches are to be fully wired with appropriate wire and terminal contacts.

Contact voltage drop: The contact voltage drop with two terminals and the switch contact in series shall not exceed 8 millivolts. This measurement shall be made from one wire contact through the switch contacts to the other wire contact with .1 ampere at a voltage of 2-4 volts DC.

Fluid immersion: Two additional qualification & group B switches, fully wired, shall be subjected to three exposure cycles in accordance with a and b.

a. The terminal end of the switch shall be immersed to a depth of 3/8-inch measured 2 +1/2, -0 minutes with a maximum of two minutes between immersions. After each immersion the excess liquid is to be blown off the switch external surfaces with an air jet.

(1) MIL-T-5624 - Turbine fuel, aviation, grade JP-4 or JP-5.

(2) Skydrol 500A - Federal stock number 9150-857-9069.

(3) Coolanel - Federal stock number 9150-551-4022.

(4) MIL-E-9500 - Ethylene glycol, technical uninhibited.

(5) *MIL-L-7808 - Lubricating oil, aircraft turbine engine synthetic base.*

b. Exposure to ambient air for 24 ±2 hours.

c. At the end of the third cycle, the insulation resistance shall be measured and the switches shall be inspected for cracking and loosening of bonds and seams. When switches are tested as specified the insulation resistance shall not be less than 1,000 megohms and there shall be no evidence of cracking and loosening of bonds and seams.

User activities: Army - MI
Navy -
Air Force -

Review activities: Army - AR, AV
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Procurement Specification MIL-S-8834		SUPERSEDES: MS0024656	PAGE 5 OF 5

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